

Astronomers often say that large telescopes are like time machines, or equivalently, that distant light is old light, because of the finite speed at which light travels.

Although we cannot tell that light is not instantaneous in our everyday life, when traveling through the immense distances in space it can no longer be thought of that way. As light travels at a constant and finite speed, "lookback time" refers to looking at objects far away as they existed in the past. When astronomers detect a supernovae, explosion occurring in a galaxy at a distance of 100 million light years, this explosion did not happen this evening, but rather 100 million years ago, as it took 100 million years for the light from the explosion to travel to our telescopes.

Through the use of more and more powerful telescopes allowing us to view objects farther and farther away we are looking back deeper and deeper into the history of the universe, and studying the formation of the earliest galaxies and a look at the history of the universe soon after the Big Bang.

The evidence that the universe began in a hot, dense state 13.8 billion years ago is found in the expansion of space, the afterglow of the Big Bang, and the observable cooling of the cosmos.

These are several major points which are in favor of the Big Bang concept: Hubble's Law and redshift: observations show that every galaxy, almost without exception, is moving away from us; its light is shifted towards the red end of the spectrum. And, galaxies further away are moving faster: this relation between velocity and distance is called the Hubble Constant. If this expanding Universe "reverts back", all matter and energy were confined to an infinitely small point at infinitely high temperatures and energies. Cosmic Microwave Background (CMB): this is the most important piece of evidence for a hot Universe. The CMB radiation consists of very uniform light which is released at the time where light began to be freely able to travel space without being re-absorbed. This would have been a glowing hot intense radiation of extremely high energy; but as the universe expands, it is stretched out into the microwave spectrum we observe today. Temperature: The universe cools as it expands, and the Big Bang model predicts

this perfectly. It accurately predicts the temperature of the Universe today, and also the proportions of the lightest elements that we can see today (hydrogen and helium), because they could only be produced at these high temperatures.

The two dominant ingredients of the universe are dark energy and dark matter, and astronomers are unsure of their physical nature because neither interacts directly with light.

So, the stars, and we, are made of 5% of the matter of the universe. What about the other 95%? They can be divided into two further mysteries.

1. Dark Energy (68%): There is a repulsive force which seems to permeate space. Experiments with distance supernovae revealed that the rate of expansion of the universe is not actually decelerating due to the presence of gravity, but in fact, accelerating. Dark Energy is the label we put to "energy of space" that causes this acceleration.

2. Dark Matter (27%): This is an invisible form of mass which appears to provide a scaffolding for the universe. We only know of this matter through observing how galaxies behave. For example, objects on the extremities of a galaxy orbit the center of it much too quickly to be explained solely by the visible matter, which requires enormous amounts of hidden mass.

Neither of these things have been determined by astronomers because we don't know what they're made of, and cannot see them using the many forms of electromagnetic waves that our current technology uses for observation, from radio waves to gamma rays. The reason we know they are there is because of their overwhelming effect on visible matter gravitationally. Dark Energy is not a collection of particles we can recreate in a laboratory.